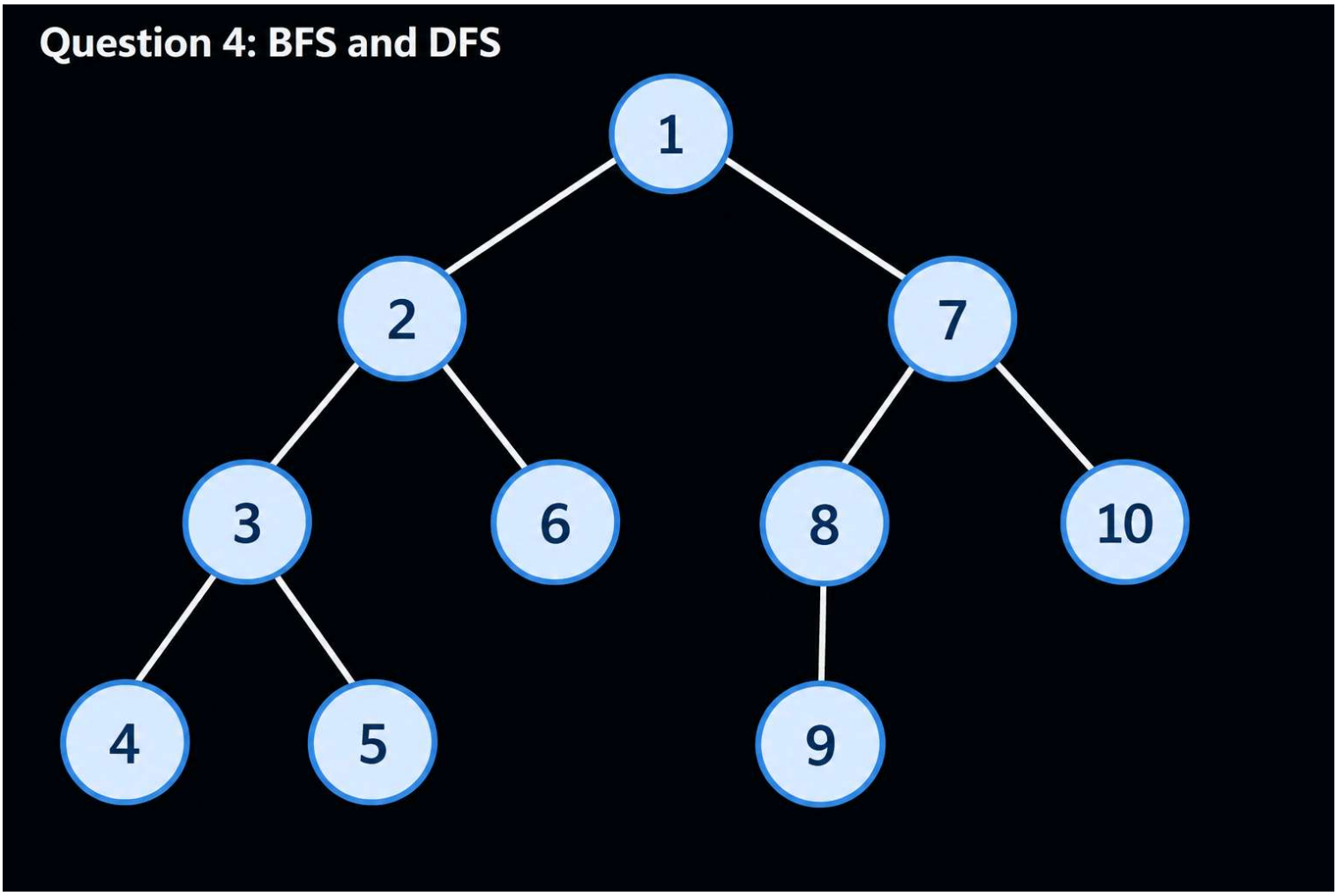
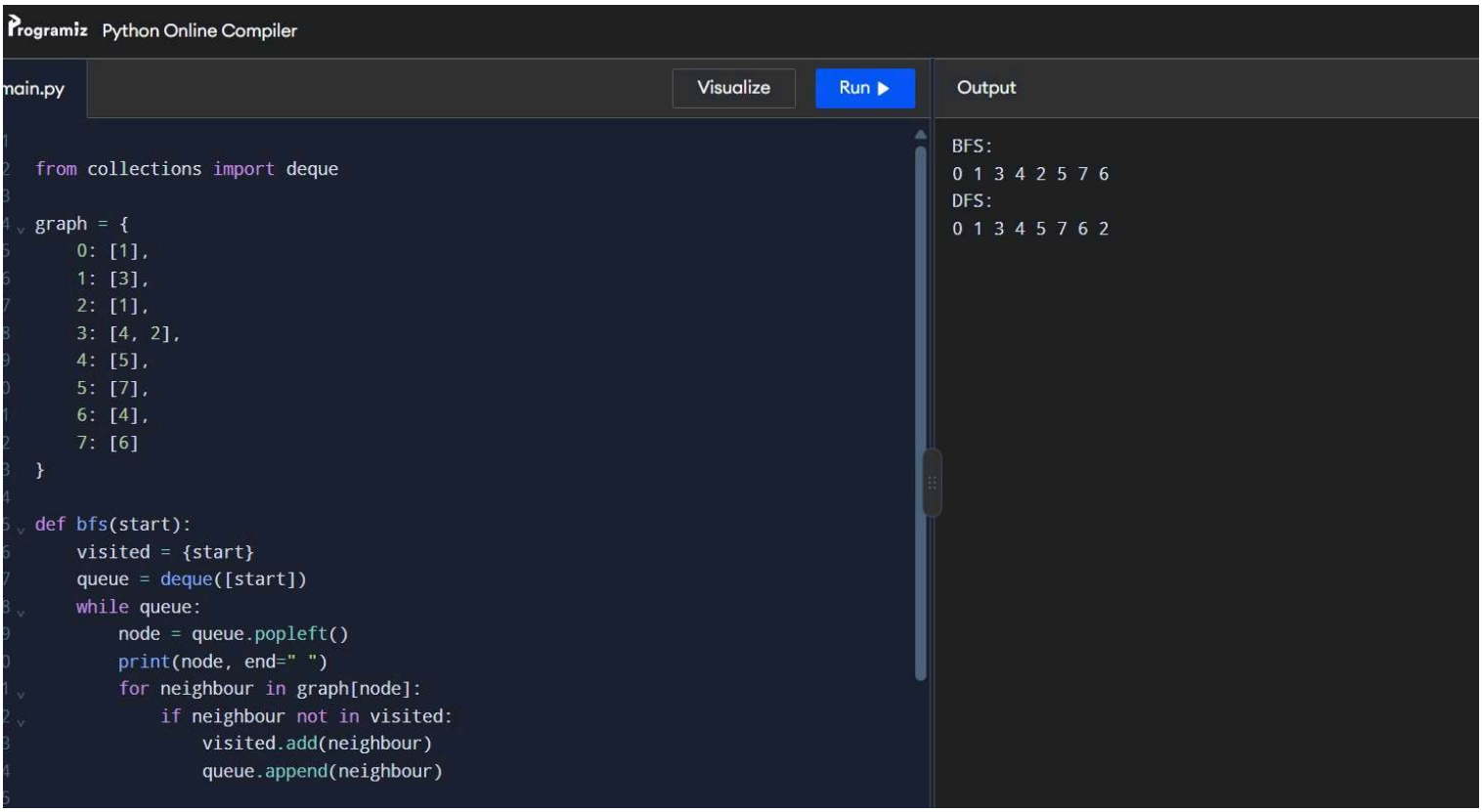
Question 4: BFS and DFS




Programiz Python Online Compiler
main.py
Visualize
Run
Output
from collections import deque
graph = {
0: [1],
1: [3],
2: [1],
3: [4, 2],
4: [5],
5: [7],
6: [4],
7: [6]
}
def bfs(start):
visited = {start}
queue = deque([start])
while queue:
node = queue.popleft()
print(node, end=" ")
for neighbour in graph[node]:
if neighbour not in visited:
visited.add(neighbour)
queue.append(neighbour)
BFS:
0 1 3 4 2 5 7 6
DFS:
0 1 3 4 5 7 6 2

```
def dfs(node, visited):
    visited.add(node)
    print(node, end=" ")
    for neighbour in graph[node]:
        if neighbour not in visited:
            dfs(neighbour, visited)

print("BFS:")
bfs(0)

print("\nDFS:")
dfs(0, set())
print()
```